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FINAL PROJECT REPORT

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1. INTRODUCTION

1.1 Project Overview

The "Auto Ticket Classification using Flow Designer" project aims to revolutionize traditional IT Service Management (ITSM) workflows. By integrating cloud-native orchestration tools with Machine Learning, this project automates the triage, categorization, and routing of inbound IT support tickets based on natural language descriptions.

1.2 Purpose

The primary purpose is to drastically reduce Mean Time to Resolution (MTTR) by eliminating the manual dispatch process. Automated classification ensures that tickets reach the correct resolver group instantly, freeing up L1 support agents to focus on complex problem-solving rather than administrative routing.

2. IDEATION PHASE

2.1 Problem Statement

> **Problem:** Manual triage and routing of IT support tickets is time-consuming, prone to human error, and significantly delays resolution times for critical technical issues. End users describe problems using free-text, natural language that lacks standard terminology, causing static rules to fail.

2.2 Empathy Map Canvas

IT Support Agents (Users): Feel overwhelmed by ticket backlogs and frustrated by repetitive reading of vague descriptions. They need an efficient way to sort incidents.
End-Users (Customers): Feel delayed and annoyed when their tickets bounce between incorrect departments. They want fast, accurate support.

2.3 Brainstorming

During ideation, the team evaluated multiple approaches: keyword-based static routing, hiring more dispatchers, or utilizing AI. We settled on using **ServiceNow Predictive Intelligence** combined with **Flow Designer** because it leverages historical data to understand context, which scales better than static keywords and is cheaper than manual labor.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- **Awareness/Trigger:** User experiences a technical issue (e.g., "My email won't sync").
- **Action:** User logs into the Service Portal and types a brief, natural language description of the problem.
- **System Interaction:** The system captures the text, analyzes it in real-time, predicts "Category: Email, Priority: High", and triggers a workflow.
- **Resolution:** The ticket appears directly in the Network/Exchange Team's queue, and the user receives instant confirmation of assignment.

3.2 Solution Requirement

The solution requires a highly available cloud architecture, robust Natural Language Processing (NLP) capabilities, and an integration layer capable of orchestrating ITSM logic without custom coding (low-code/no-code approach). Data privacy controls (ACLs) are mandated to protect user information.

3.3 Technology Stack

Component	Technology Used
User Interface	Service Portal (HTML, CSS, AngularJS)
Application Logic	ServiceNow Flow Designer (Low-code workflow)
Machine Learning	Predictive Intelligence / Python (spaCy/NLTK)
Database	Cloud-native PaaS Database (MariaDB/MySQL)
Infrastructure	Enterprise SaaS Cloud Platform

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution perfectly fits the enterprise constraints. By replacing rigid keyword scripts with an intelligent ML model, the solution adapts to how users actually speak. The implementation of Flow Designer allows process owners to modify the routing logic visually without deep programming knowledge.

4.2 Solution Architecture

The architecture bridges the gap between end-user input and database record creation. The flow is as follows:

1. **End User** submits a ticket via the Web UI.
2. **Flow Designer** intercepts the creation event and extracts the text.
3. **Predictive Intelligence (ML)** processes the text and returns confidence scores for categories.
4. **Flow Designer** applies logic (e.g., if confidence > 80%, assign automatically).
5. **Incident DB** updates the record and notifies the assignee.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed using Agile methodologies, divided into Epics and Sprints.

- **Sprint 1 (12 Story Points):** Focused on Data Setup & UI. Tasks included configuring the Service Portal, extracting historical incident data, and cleaning text for model training.
- **Sprint 2 (20 Story Points):** Focused on ML Integration and Flow Logic. Tasks included training the classification model, configuring the Flow Designer routing steps, and end-to-end testing.

> **Velocity Calculation:** Total Points (32) / Number of Sprints (2) = Average Velocity of **16 Points per Sprint**.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Test Cases & UAT

During User Acceptance Testing (UAT), the system demonstrated high reliability. A total of 77 defects were logged and resolved (37 fixed, 20 closed by design). Critical test areas passed included:

- Classification Engine: 7/7 passed.
- ServiceNow Portal UI: 51/51 passed.
- Flow Designer Routing Logic: 3/3 passed.

7. RESULTS & CONCLUSION

7.1 Advantages & Disadvantages

Advantages:

- **Speed:** Instantaneous ticket routing drastically reduces MTTR.
- **Cost Efficiency:** Lowers Operational Expenditure (OpEx) by reducing the need for L1 dispatch personnel.
- **Scalability:** Cloud-native flow can handle massive spikes during major outages.

Disadvantages:

- Requires a large volume of historical, accurately categorized data to train the ML model effectively.
- May misclassify highly ambiguous or entirely novel technical issues (Edge cases).

7.2 Conclusion

The integration of ServiceNow Flow Designer with Machine Learning for Auto Ticket Classification successfully achieved its objective. It replaced manual, error-prone triage with a seamless, intelligent automated pipeline. The business impact includes faster resolution times, happier end-users, and more efficient IT staff allocation.

8. FUTURE SCOPE

Future iterations of this project could include:

- **Multi-lingual NLP Support:** Enabling auto-classification for global teams submitting tickets in different languages.
- **Self-Healing IT:** Linking the classification engine directly to automated remediation scripts (e.g., auto-resetting a password or clearing a cache without any human agent involvement).

9. APPENDIX

- **Source Code / Configurations:** [Update Sets retained in ServiceNow instance repository]
- **Dataset Link:** [Anonymized ITSM Historical Data v1.2]
- **GitHub & Project Demo:** https://github.com/John-glacier/Auto-Ticket-Classification-using-Flow-Designer.git